Fig. 2: Adaboost decision matrix algorithm (Courtesy: Google Images)

trauma that the survivors of the accidents have to go through. This spills into other things as well, such as huge medical bills, high insurance rates, damage or loss of property, etc.

It is needless to say that we need autonomous vehicles. But apart from the risk to life, there are other interesting reasons as well. Let us briefly look through some of those.

Why we need autonomous vehicles

Besides saving human lives, there are other major benefits of autonomous vehicles as well. Such as:

1. Self-driving cars can help people with disabilities. Imagine being blind and still being able to own a car and go anywhere.

2. Self-driving cars mean less accidents. Which in turn means no need to pay a high premium on your insurance.

3. More than half of the traffic jams occur due to traffic flow disruptions which are caused primarily by fender-benders and other accidents. With autonomous vehicles, the time spent on the road can massively decrease.

4. You can save money on fuel each year as fewer traffic disruptions help you commute faster, thus using less fuel than you do now. This also helps the environment.

5. Self-driving vehicles reduce the cost of logistics and will make delivery of products to your doorstep much faster and cheaper.

6. In the pandemic that we are in right now, delivery of essentials has been a challenge as it leads to inevitable contact between you and the delivery executive. Additionally, it also puts the life of the delivery executive at risk. By using autonomous vehicles, social distancing during epidemics/pandemics can be maintained better, thus saving more lives.

7. Emergencies such as curfews, outbreak of diseases or even war often cause people to get stranded. Delivery of essentials such as medicine and food to such places is an ongoing challenge. With the help of self-driving vehicles, much needed assistance can be provided in times of emergencies.

How does a self-driving car work?

There are two aspects to self-driving cars that we should consider in order to understand, as simply as possible, how a self-driving car works. The first aspect to consider is the hardware that goes into an autonomous vehicle. The second is the software that runs in the background making the car functional.

The hardware required for an autonomous vehicle includes the following:

1. Lidar (light detection and ranging)
2. Radar
3. Sophisticated cameras
4. Integrated photonics circuit
5. Mach-Zender modulator
6. Main computer

The software, languages, frameworks, and tools used to build an autonomous vehicle include:

1. MATLAB
2. Simulink
3. Version control system (Git)
4. Vehicle design—CAE/CAD/PLM/FEA
5. LS-DYNA, MSC ADAMS, CarSim, CarMaker, Dymola, OptimumG, SusProg3D, Oktal SKANeR, etc.
6. Docker, CMake, Shell, Bash, Perl, JavaScript, Node.js, React, Go, Rust, Java, Redux, Scala, R, Ruby, Rest API, gRPC, protobuf, Julia, HTML5, PHP
7. Verilog, VHDL, DSP, Cadence, Synopsys, Xilinx Platform Studio (ISE and XPS)

Apart from the list of software and tools mentioned above, the data is stored in the cloud and the control algorithms are trained on highly sophisticated deep learning and machine learning models. How it all comes together?

The car has a main computer, generally located in the back trunk, that controls all the necessary software installed. It collects data from all the sensors, processes it, and decides using highly accurate deep learning models. The computer learns with every decision it makes and gets better and better the more the car is driven. The computer also requires a high-quality and low-latency connection to the Internet, ideally using 5G technology.

The radar sensors placed strategically around the car, as shown in Fig. 1, help in detecting objects and obstacles on the road. Additionally, they help measure the distance of the car from the obstacles. This is highly important for avoiding hitting immobile objects or pedestrians and animals. It also facilitates changing lanes, taking turns, and adjusting the speed.

But detecting the existence of an object isn't enough for the car to drive safely. You require a technology that can not only detect an object and its distance from you, but also its shape and density in order to identify what the object is. Why is this important?